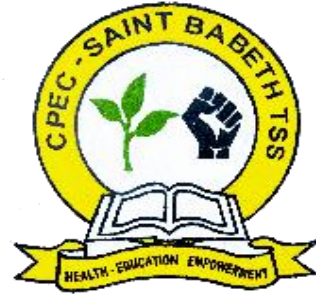


MODULE CODE: SWDBD401 BACKEND
APPLICATION DEVELOPMENT

Date: 5th November, 2025

Period: 08H00:30 AM- 11H00 :30AM



SCHOOL BASED TERM ONE EXAMINATION

SECTOR: ICT AND MULTIMEDIA

TRADE: SOFTWARE DEVELOPMENT

RTQF LEVEL: 4

MARKS:

DURATION: 3 HOURS

TRAINEE'S FULL NAMES:

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed of Three Sections (A, B, C). Follow the instructions given below, and answer the indicated questions for a total of 100marks

Section A: Fifteen (15) questions, all Compulsory. 55 marks

Section B: Five (5) questions, Attempt any 3 Questions of your choice. 30 marks

Section C: Two (2) questions, Attempt any 1 Question of your choice. 15 marks

SECTION A: ATTEMPT ALL QUESTIONS**(55 Marks)**

1. Define the following terms as used in Backend Development with Node.js:

i. Object:

.....
..... **/2marks**

ii. Routes:

.....
..... **/2marks**

iii. Dependencies:

.....
..... **/2marks**

iv. Postman:

.....
..... **/2marks**

v.

Nodemon:.....
..... **/2marks**

2. What is Node.js? **/2.5marks**

- a) A front-end JavaScript framework
- b) A JavaScript runtime built on Chrome's V8 engine
- c) A database management system
- d) A package manager for JavaScript

3. What is data encryption? **/2.5marks**

- a. The process of compressing data to save storage space
- b. The process of converting readable data into an unreadable format to prevent unauthorized access
- c. The process of backing up data for future use
- d. The process of deleting sensitive data permanently

4. Which HTTP method is typically used to retrieve data from a server in a RESTful API? **/2.5marks**

- a) POST
- b) PUT
- c) GET
- d) DELETE

5. The primary purpose of data encryption is to: **/2.5marks**

- a) Compress data to save space
- b) Convert data into an unreadable format to prevent unauthorized access
- c) Speed up data transmission
- d) Create backups of sensitive information

6. Which tool automatically restarts a Node.js application when file changes are detected, improving the development workflow? **/2.5marks**

13. Fill in the blanks with the correct terms from the list below: Terms to Use: Node.js, AES, GET /**6marks**

a.....is a symmetric encryption algorithm used for fast data encryption

b. The method that is used to retrieve data from a server is called.....

c. Express.js is a popular web application framework for.....

14. Write a code that create a simple express sever with express.js to display

“Welcome To CPEC Saint Babeth Technical Secondary School” End

point which runs on port on <http://localhost:3500> as response to the browser. /**4marks**

Section B: Attempt any three (3) questions (30 marks)

15. Explore any ten(10) advantages of the Express.js web application framework.

16. Write a Node.JS application to calculate addition and average of four numbers and display the results in the console.

17. Define Express.js and its installation process.

18. Match the HTTP method in Column X with its typical usage in a RESTful API in Column Y. /**15marks**

Column X (HTTP Method)	Column Y (Typical Usage)	Answers
1. GET	A. Delete a resource	1.....
2. POST	B. Update or replace a resource	2.....
3. PUT	C. Retrieve data	3.....
4. DELETE	D. is a request method designed for making partial modifications to an existing resource on a server	4.....
5. PATCH	E. Create a new resource	5.....

Section C: Attempt only one (1) question (15 marks)

19. Fill the followings with correct key word from the listed one below (POST, end, npm install mysql, 500 Internal Server Error, Postman, Logging, 404 Not Found, 200 OK, Error Handling, Create, Read, Update, and Delete, PUT, listen, meets, 201 Created, MySQL Connection, GET, body-parser, validate)

- i. To create a new resource, you typically use the HTTP method.
- ii. The command to install the MySQL package in a Node.js project is.....
- iii. When a server encounters an unexpected issue, it typically responds with a status code.
- iv. To test API endpoints, you can use a tool called
- v. The middleware is responsible for logging the details of incoming requests.
- vi. A resource that does not exist in the database should return status code.
- vii. The method used to update an existing resource is typically.....
- viii. An Express server can be started by calling the method on the app object.
- ix. A successful resource creation in a RESTful API returns a status code.
- x. The method is used to request data from a specified resource.

20. ABC LTD Company would like to develop a Library Management System backend using Node.js. The system will allow the library staff to manage books and users. You are hired and tasked to create a project folder on the desktop. Describe all the steps you will pass through.

MERRY CHRISTMAS AND HAPPY NEW YEAR OF 2026!!!